Google API Key and CX Setup Guide

Ghost Identity Hunter Team

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> **Important Note**
>
> This guide explains how to set up Google Custom Search API credentials for use with Ghost Identity Hunter's Google Dorks feature. The Google Custom Search API requires a billing account for API access.

Contents

1 Overview

Google Custom Search API allows Ghost Identity Hunter to perform advanced username discovery using Google Dorks without rate limiting. This guide covers:

- Creating a Google Cloud project
- Enabling Custom Search API
- Creating API credentials
- Setting up Custom Search Engine
- Configuring environment variables
- Testing the setup

2 Prerequisites

Before starting, ensure you have:

- A Google account
- Access to Google Cloud Console
- A billing account (required for API access)
- Basic familiarity with web interfaces

3 Step 1: Create Google Cloud Project

3.1 Access Google Cloud Console

1. Navigate to `https://console.cloud.google.com/`
2. Sign in with your Google account

3.2 Create New Project

1. Click the project dropdown in the top navigation bar
2. Click **"New Project"**
3. Enter a project name (e.g., "Ghost Identity Hunter")
4. Click **"Create"**
5. Wait for project creation to complete

4 Step 2: Enable Custom Search API

4.1 Navigate to API Library

1. In the left sidebar, navigate to **APIs & Services** → **Library**
2. Search for "Custom Search API"
3. Click on **Custom Search API** from the results

4.2 Enable the API

1. Click the **"Enable"** button
2. Wait for the API to be enabled
3. You should see "API enabled" confirmation

Billing Requirement

The Custom Search API requires a billing account to be linked to your project. If you don't have one, you'll be prompted to add billing information during this step.

5 Step 3: Create API Key

5.1 Navigate to Credentials

1. In the left sidebar, navigate to **APIs & Services** → **Credentials**
2. Click the **"Create Credentials"** button
3. Select **"API Key"**

5.2 Configure API Key

1. Copy the generated API key
2. Click on the API key to edit its settings
3. Under **Application restrictions**, select **"None"** for testing
4. Under **API restrictions**, select **"Custom Search API"**
5. Click **"Save"**

API Key Format

Your API key will look like: `AIzaSyBgd95vN5dUOMTpffIdYiP8WJRhcQe6_VA`

6 Step 4: Create Custom Search Engine

6.1 Access Programmable Search

1. Navigate to `https://programmablesearchengine.google.com/`
2. Sign in with your Google account

6.2 Create New Search Engine

1. Click **"Add"** to create a new search engine
2. **Sites to search**: Enter a valid domain (e.g., `google.com`)
3. **Name**: Enter a name (e.g., "Ghost Identity Hunter")
4. Click **"Create"**

6.3 Configure Search Engine

1. After creation, click **"Control Panel"** for your search engine
2. Go to the **Setup** tab
3. Under **Advanced** settings:
   - Enable **"Search the entire web"**
   - Set **"Safe Search"** to **OFF**
4. Click **"Save"**

6.4 Get CX ID

1. In the **Setup** tab, look for **"Search engine ID"**
2. Copy the CX ID

CX ID Format

Your CX ID will look like: `22b3a2bb157154ea2`

7 Step 5: Configure Environment Variables

7.1 Add to Shell Configuration

Add the following lines to your shell configuration file:

- **macOS/Linux**: `~/.zshrc` or `~/.bashrc`
- **Windows**: System Environment Variables

```
export GOOGLE_API_KEY="your_actual_api_key_here"
export GOOGLE_CX="your_actual_cx_id_here"
```

7.2 Reload Configuration

```
# For macOS/Linux with zsh
source ~/.zshrc

# For macOS/Linux with bash
source ~/.bashrc

# For Windows, restart Command Prompt or PowerShell
```

7.3 Verify Configuration

```
echo $GOOGLE_API_KEY
echo $GOOGLE_CX
```

8 Step 6: Test the Setup

8.1 Test API Directly

```
curl "https://www.googleapis.com/customsearch/v1?key=$GOOGLE_API_KEY&cx
   =$GOOGLE_CX&q=test&num=1"
```

Expected response: JSON with search results.

8.2 Test with Ghost Identity Hunter

```
python -m src.cli investigate --username "test_user" --use-google-dorks
    --use-google-api
```

Expected: Investigation runs without 403 or 429 errors.

9 Troubleshooting

9.1 Common Issues

Error	Solution
403 Forbidden	Enable Custom Search API in Google Cloud Console
403 Forbidden	Add billing account to your project
403 Forbidden	Check API key restrictions (set to "None" for testing)
403 Forbidden	Ensure CX ID and API key are from same project
429 Too Many Requests	Use API instead of web scraping
Invalid CX ID	Verify CX ID in Programmable Search Control Panel
Invalid API Key	Check API key in Google Cloud Console

Table 1: Common Error Messages and Solutions

9.2 Verification Checklist

- ✓ Custom Search API is enabled in Google Cloud Console
- ✓ Billing account is linked to the project
- ✓ API key has correct restrictions
- ✓ CX ID is valid and "Search the entire web" is enabled
- ✓ Environment variables are set correctly
- ✓ Shell configuration has been reloaded
- ✓ Direct API test returns JSON results

10 Usage Examples

10.1 With Environment Variables

Once configured, use Google Dorks without specifying credentials:

```
python -m src.cli investigate --username "target_user" --use-google-
   dorks --use-google-api
```

10.2 With Explicit Credentials

Override environment variables with explicit values:

```
python -m src.cli investigate --username "target_user" \
  --use-google-dorks \
  --use-google-api \
  --google-api-key "YOUR_API_KEY" \
  --google-cx "YOUR_CX_ID"
```

10.3 Without Google Dorks

If API setup fails, the application works without Google Dorks:

```
python -m src.cli investigate --username "target_user"
```

11 Security Best Practices

- Never commit API keys to version control
- Use environment variables for sensitive credentials
- Restrict API key usage to specific applications in production
- Monitor API usage in Google Cloud Console
- Rotate API keys periodically
- Use separate projects for development and production

12 Additional Resources

- https://console.cloud.google.com/ - Google Cloud Console
- https://programmablesearchengine.google.com/ - Programmable Search
- https://developers.google.com/custom-search/v1/overview - API Documentation
- https://cloud.google.com/docs/authentication/api-keys - API Key Best Practices

13 Summary

Setting up Google Custom Search API involves:

1. Creating a Google Cloud project
2. Enabling Custom Search API (requires billing)
3. Creating and configuring API key
4. Setting up Custom Search Engine with web-wide search
5. Configuring environment variables
6. Testing the setup

Once configured, Ghost Identity Hunter can perform unlimited Google Dorks searches without rate limiting, providing more comprehensive OSINT results.

Alternative

If you cannot set up Google API credentials, Ghost Identity Hunter still works perfectly without Google Dorks. The core OSINT features (email analysis, username search, platform detection, graph correlation) all function through other data sources.